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1. Genetic Algorithm Application

Fitness Value Calculation

The fitness function is given as $f(x) = x^2$. We calculate the fitness for each individual in the initial population by squaring their value:

- **Individual 2:** $f(2) = 2^2 = 4$
- **Individual 4:** $f(4) = 4^2 = 16$
- **Individual 6:** $f(6) = 6^2 = 36$
- **Individual 8:** $f(8) = 8^2 = 64$

Selection

In a Genetic Algorithm, the candidates with the highest fitness scores are selected for reproduction. Based on the calculated scores, the two best candidates are **6** (fitness: 36) and **8** (fitness: 64).

Crossover Operation

To perform crossover, we first convert the selected individuals into binary representation (using 4 bits):

- $6 = 0110_2$
- $8 = 1000_2$

The problem asks to exchange the "least significant bits".

- **If exchanging the single Least Significant Bit (LSB):** The LSB for both 0110 and 1000 is 0. Exchanging them results in no change. The offspring remain 0110 (6) and 1000 (8).
- **If interpreting as a 1-point crossover exchanging the *two* least significant bits:**
  - Parent 1 $(01|10)$ donates 10
  - Parent 2 $(10|00)$ donates 00
  - **Offspring 1:** $0100_2 = 4$
  - **Offspring 2:** $1010_2 = 10$

Role of Mutation

Mutation introduces random, small alterations to an individual's genetic sequence (e.g., flipping a 0 to a 1 in a binary string). Its primary role is to maintain **genetic diversity** within the population. By randomly exploring new areas of the search space, mutation prevents the algorithm from converging too early on a "local optimum" (a solution that is good, but not the absolute best possible solution) and ensures the algorithm remains effective over multiple generations.

2. Perceptron Classification

The Given Parameters

- **Inputs:** Advertisement exposure ($x_1 = 2$), Income level ($x_2 = 1$)
- **Weights:** $w_1 = 0.5$, $w_2 = 0.3$
- **Bias:** $b = -0.4$

Computing the Net Input

The net input (Z) to a perceptron is the sum of the weighted inputs plus the bias. The formula is:

$$Z = (x_1 \cdot w_1) + (x_2 \cdot w_2) + b$$

Plugging in the values:

$$Z = (2 \cdot 0.5) + (1 \cdot 0.3) + (-0.4)$$

$$Z = 1.0 + 0.3 - 0.4$$

$$Z = 0.9$$

Applying the Step Activation Function

A standard step activation function outputs 1 if the net input is greater than or equal to 0, and outputs 0 if the net input is less than 0.

- Since our net input $Z = 0.9$, and $0.9 \geq 0$, the output of the activation function is **1**.

Conclusion

Because the perceptron outputs a 1, the model predicts that the customer **will make a purchase**.